NAME : Dakshata Chandrakant Salgaonkar

Div : SE3-B2

Roll No: 30

Branch : CSE-DS

Experiment No : 3

Experiment Title : Design and Implementation of a Dynamic To-Do List Using JavaScript and DOM

Aim : To design and develop a dynamic To-Do List application using JavaScript and DOM manipulation that allows users to add, mark as completed, and delete tasks efficiently.

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Smart To-Do List</title>


  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: "Segoe UI", sans-serif;
    }

    body {
      background: linear-gradient(135deg, #667eea, #764ba2);
      height: 100vh;
      display: flex;
      align-items: center;
      justify-content: center;
    }
```

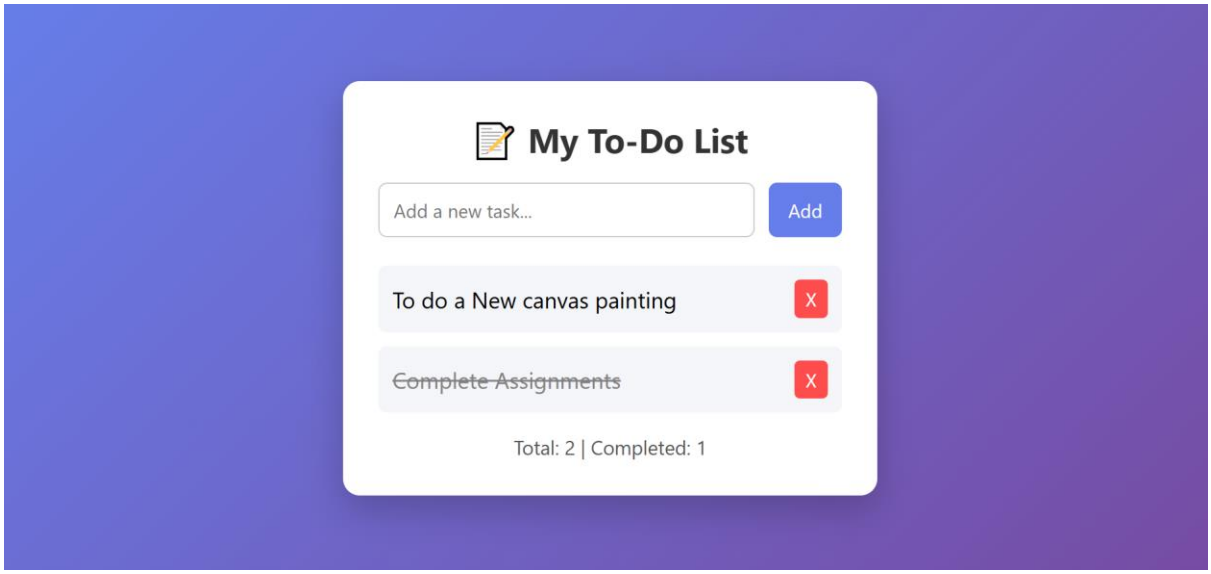
```
.todo-app {
  background: #fff;
  width: 380px;
  padding: 25px;
  border-radius: 12px;
  box-shadow: 0 10px 25px rgba(0,0,0,0.2);
}

h2 {
  text-align: center;
  margin-bottom: 15px;
  color: #333;
}

.input-group {
  display: flex;
  gap: 10px;
}

input {
  flex: 1;
  padding: 10px;
  border-radius: 6px;
  border: 1px solid #ccc;
  outline: none;
}

button.add-btn {
  background: #667eea;
  color: white;
  border: none;
```

```
  padding: 10px 14px;
  border-radius: 6px;
  cursor: pointer;
}

button.add-btn:hover {
  background: #5a6bdc;
}

ul {
  list-style: none;
  margin-top: 20px;
}

li {
  background: #f5f6fa;
  padding: 10px;
  border-radius: 6px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 10px;
  transition: 0.3s;
}

li.completed span {
  text-decoration: line-through;
  color: gray;
}

li span {
```

```
      cursor: pointer;
      flex: 1;
    }

    .delete {
      background: #ff4d4d;
      border: none;
      color: white;
      padding: 5px 8px;
      border-radius: 4px;
      cursor: pointer;
    }

    .delete:hover {
      background: #e84141;
    }

    .stats {
      margin-top: 15px;
      text-align: center;
      font-size: 14px;
      color: #555;
    }
  </style>
</head>

<body>

  <div class="todo-app">
    <h2>📝 My To-Do List</h2>
```

```
  <div class="input-group">
    <input type="text" id="taskInput" placeholder="Add a new task...">
    <button class="add-btn" onclick="addTask()">Add</button>
  </div>

  <ul id="taskList"></ul>

  <div class="stats" id="stats"></div>
</div>

<script>
  const taskInput = document.getElementById("taskInput");
  const taskList = document.getElementById("taskList");
  const stats = document.getElementById("stats");

  let tasks = JSON.parse(localStorage.getItem("tasks")) || [];

  function renderTasks() {
    taskList.innerHTML = "";
    tasks.forEach((task, index) => {
      const li = document.createElement("li");
      li.className = task.completed ? "completed" : "";

      li.innerHTML = `
        <span onclick="toggleTask(${index})">${task.text}</span>
        <button class="delete" onclick="deleteTask(${index})">X</button>
      `;

      taskList.appendChild(li);
    });
```

```
  updateStats();
  localStorage.setItem("tasks", JSON.stringify(tasks));
}

function addTask() {
  const text = taskInput.value.trim();
  if (text === "") return alert("Enter a task!");

  tasks.push({ text, completed: false });
  taskInput.value = "";
  renderTasks();
}

function toggleTask(index) {
  tasks[index].completed = !tasks[index].completed;
  renderTasks();
}

function deleteTask(index) {
  tasks.splice(index, 1);
  renderTasks();
}

function updateStats() {
  const completed = tasks.filter(t => t.completed).length;
  stats.textContent = `Total: ${tasks.length} | Completed: ${completed}`;
}

taskInput.addEventListener("keypress", function(e) {
  if (e.key === "Enter") addTask();
});
```

```
  renderTasks();
 </script>


</body>
</html>
```

Output: